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How reliably can an NLP model trained on social media emotions identify fine-grained emotions in customer support messages?

Team 1

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Word Count: 4000

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Member	Contribution
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Abstract

Customer support interactions are an underexplored domain for fine-grained emotion detection, yet understanding customer emotions has significant business value. This paper investigates how reliably a transformer-based NLP model, fine-tuned on the GoEmotions dataset (Reddit), can identify emotions in real customer support messages from Twitter. We fine-tuned BERT-base-uncased on GoEmotions and evaluated cross-domain transfer against a manually annotated sample of 500 customer support tweets ($\kappa = 0.733$).

BERT achieved a weighted F1 of 0.566 in-domain but dropped to 0.312 on Twitter, a transfer gap of 25.5 percentage points, constituting poor transfer by our pre-defined thresholds. A TF-IDF Logistic Regression baseline exhibited a near-identical gap (26.0pp), suggesting the bottleneck is domain mismatch rather than model sophistication. Collapsing the 28 fine-grained classes into three broader groups (negative, positive, ambiguous) improved Twitter weighted F1 to 0.485, indicating that coarse emotional triage transfers more robustly. We recommend that organisations avoid deploying fine-grained emotion labels without domain-specific adaptation and instead prioritise collecting in-domain labelled data or adopt grouped emotion labels for routing and escalation.

1. Introduction

Customer support interactions are fundamentally emotional exchanges. The ability to automatically detect specific emotions, distinguishing confusion from anger, or disappointment from annoyance, could enable smart ticket routing, personalised automated responses, and pre-emptive escalation of service inquiries. Research demonstrates that customer emotions strongly predict service quality perceptions and overall satisfaction (Mattila and Enz, 2002). However, most deployed systems rely on crude positive/negative/neutral sentiment classification or keyword-based routing that bypasses emotional context entirely.

Recent advances in transformer-based architectures have enabled models to classify several emotional categories with good accuracy (Harutyunyan and Kumar, 2026). However, most of these models are trained on general social media content (Kaur and Sharma, 2022), leaving a research gap: can emotion classifiers trained on casual social media text generalise to the context-dependent, task-focused domain of customer service?

This paper addresses that gap by asking: “How reliably can an NLP model trained on social media emotions (GoEmotions/Reddit) identify fine-grained emotions in customer support messages (Twitter)?” We fine-tuned a BERT-based classifier on the GoEmotions dataset (Demszky et al., 2020), comprising 58,000 Reddit comments across 28 emotion categories, and applied it to real customer support interactions from Twitter (ThoughtVector, 2017). We examine performance quantitatively at both aggregate and per-emotion levels, analyse errors through systematic qualitative examination of misclassified tweets, and conclude with actionable recommendations for the customer service industry.

2. Related Works

Emotion detection is a well-established NLP task extending beyond binary sentiment analysis to identify nuanced categories such as anger, joy, fear, and gratitude. Early lexicon-based methods matched text against manually tagged word lists, assigning emotion scores based on keyword presence. While computationally efficient, these approaches struggled with informal

language where meaning depends heavily on surrounding context, and were unable to handle negation, sarcasm, or figurative language common in social media (Muhammad, 2016). Classical supervised methods such as support vector machines improved upon this by learning from labelled training data, but largely treated words in isolation through bag-of-words representations, failing to capture the contextual shifts in meaning that determine emotion in practice (Seyeditabari et al., 2019).

The transformer architecture (Vaswani et al., 2017) addressed this fundamental limitation through self-attention mechanisms that weigh relationships between all tokens in a sequence simultaneously, enabling models to capture long-range dependencies and contextual nuance. BERT (Devlin et al., 2019) extended this through bidirectional pre-training on large corpora, learning rich contextual representations that can be fine-tuned for downstream tasks including emotion classification. Subsequent hybrid approaches combining transformer encoders with recurrent components such as LSTMs have achieved accuracy of 88% on fine-grained emotion classification tasks (Harutyunyan and Kumar, 2026). However, it remains unclear whether these performance gains are robust to domain shift, particularly when models are applied to text whose register, vocabulary, and pragmatic conventions differ substantially from the training corpus.

The GoEmotions dataset (Demszky et al., 2020) provided the field's primary benchmark for fine-grained classification: 58,000 Reddit comments across 28 emotion categories. However, its creation revealed structural difficulties in the task itself. Inter-annotator Cohen's Kappa ranged from 0.095 for grief to 0.749 for gratitude, and Alvarez-Gonzalez et al. (2021) confirmed this reflects a fundamental ceiling on text-based emotion detection rather than mere annotation inconsistency. Semantically overlapping categories such as annoyance and anger occupy shared psychological territory that no classifier can fully resolve (Demszky et al., 2020; Truong, 2024). This has important implications for our study: even perfect domain transfer would be constrained by this inherent ambiguity.

Domain generalisation compounds this challenge. Transfer learning assumes that representations learned in one domain carry useful signal in another (Ruder, 2019), but performance degrades when target text differs in vocabulary, tone, and pragmatic function (Kumar et al., 2026). Several mitigation strategies exist in the literature. Domain-adaptive pre-training, where a model undergoes further unsupervised training on unlabelled target-domain text before task-specific fine-tuning, has shown substantial improvements for specialised domains (Gururangan et al., 2020). Zanwar et al. (2022) demonstrated that augmenting transformers with psycholinguistic features improves cross-corpus robustness, but their evaluation remained within the social media domain rather than extending to professional service contexts.

Customer support language presents a distinct challenge for emotion transfer. Messages are shorter, more polite, more transactional, and more indirect than typical social media posts. Emotion is frequently implied through event description ("my order hasn't arrived") rather than stated explicitly ("I am angry"), a pattern that Herzig et al. (2016) identified as a key challenge for NLP in customer service contexts. Despite growing commercial interest in emotion-aware customer support systems, academic evaluation of cross-domain emotion transfer to this specific context remains scarce. Our study directly addresses this gap by quantifying the transfer reliability from GoEmotions to customer support Twitter data and analysing the specific patterns of success and failure.

3. Methodology

3.1 Overview

We adopt a transfer learning paradigm: a model is trained on pre-labelled Reddit-sourced emotion data (GoEmotions) and evaluated on Twitter customer support messages. Performance disparities between the two domains quantify the model’s ability to generalise fine-grained emotion detection across contexts. The pipeline comprises five stages: (1) data acquisition, (2) preprocessing and single-label conversion, (3) model training, (4) manual annotation of the target domain evaluation set, and (5) evaluation. Two models are trained in parallel: a baseline TF-IDF Logistic Regression and a fine-tuned BERT transformer. The dual-model design allows us to isolate whether any observed transfer degradation is specific to contextual models or reflects a more fundamental domain incompatibility.

3.2 Data Sources

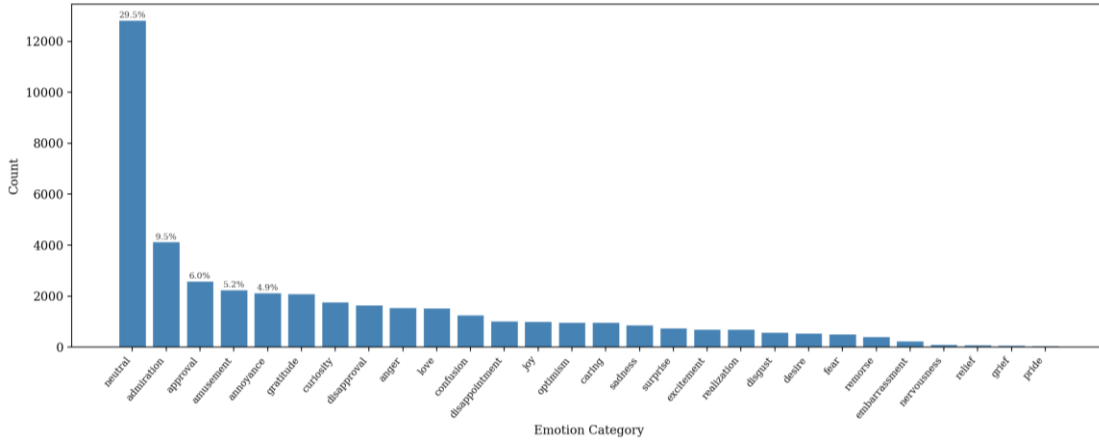
Table 1: Dataset comparison.

	GoEmotions (Training)	Twitter Customer Support (Evaluation)
Source	Reddit comments (HuggingFace)	Customer-brand interactions (Kaggle)
Reference	Demszky et al. (2020)	ThoughtVector (2017)
Total size	58,000 comments	2.8 million tweets
Used in study	Full dataset (train/val/test splits)	10,000 sampled; 500 annotated
Labels	28 emotion categories	Unlabelled (manually annotated)
Avg. text length	~20 words	~10 words
Register	Informal, expressive	Task-focused, restrained

GoEmotions is naturally multi-label, with approximately 14% of comments carrying more than one emotion label. For single-label classification, we assigned the first listed emotion per example and dropped additional labels, following standard practice in single-label GoEmotions studies (Demszky et al., 2020). We acknowledge that the ordering of labels in GoEmotions reflects the sequence in which annotators assigned them rather than a ranking of emotion dominance, which may introduce a systematic bias toward emotions that annotators tend to identify first. This is a recognised limitation of our pipeline, though its impact is mitigated by the fact that 86% of GoEmotions examples carry only a single label.

Exploratory analysis revealed significant class imbalance: neutral accounts for 29.5% of training examples, while grief, pride, and relief each represent under 1% (Figure 1). This imbalance is addressed through the use of weighted F1 as the primary evaluation metric.

Figure 1: GoEmotions emotion distribution in the training set. Neutral dominates at 29.5%.



The Twitter dataset was filtered to isolate customer-initiated messages only, yielding 1,537,843 tweets after excluding brand responses, which are typically formal and emotionally flat. From these, 10,000 tweets were randomly sampled using a fixed random seed of 42 for computational feasibility and reproducibility. Crucially, this data is entirely unseen during model training and fine-tuning, ensuring experimental validity of the transfer learning design. The substantial difference in average text length between the two datasets (approximately 20 words for Reddit versus 10 words for Twitter) reflects the structural constraints of the Twitter platform and is itself a potential source of domain mismatch.

3.3 Preprocessing

Both datasets were preprocessed consistently to ensure that observed performance differences reflect genuine domain shifts rather than artefacts of data preparation. For Twitter, @mention tags were replaced with @user to preserve syntax while removing brand noise, URLs were deleted, and whitespace was normalised.

Punctuation and capitalisation were deliberately preserved, contrary to typical NLP pipelines. Question marks can signal confusion or frustration; exclamation marks indicate emotional intensity. Retaining these features assists transformer-based emotion detection (Zanwar et al., 2022). All text was tokenised using BERT’s WordPiece tokeniser with a maximum length of 128 tokens, sufficient for both datasets given average lengths of ~20 words (Reddit) and ~10 words (Twitter).

3.4 Model Design

TF-IDF Logistic Regression (Baseline). A TF-IDF vectoriser (unigrams and bigrams, 20,000 features) paired with multinomial Logistic Regression. Bigrams capture negation phrases like “not good” where unigrams mislead. This context-free model serves as a performance floor, isolating the contribution of contextual representations.

BERT-base-uncased (Primary Model). We selected BERT-base-uncased (Devlin et al., 2019), a 12-layer transformer with approximately 109.5 million parameters. This model was chosen over alternatives for three specific reasons. First, it is the same architecture used in the original GoEmotions benchmark (Demszky et al., 2020), enabling direct and fair comparison of our results against published scores. Second, its computational requirements are feasible within our

resource constraints of a single Google Colab GPU, unlike larger models such as RoBERTa-large or DeBERTa which require substantially more memory. Third, the uncased variant is appropriate because GoEmotions text is predominantly lowercase. We also considered DistilBERT for faster training, but its distilled representations sacrifice the deeper contextual understanding needed for distinguishing semantically similar emotions such as annoyance versus disappointment. A 28-node classification head produces a probability distribution via softmax:

$$P(y = k|x) = \frac{e^{z_k}}{\sum_j e^{z_j}},$$

where z_k is the logit output for class k . The predicted class is the one with the highest probability: $\operatorname{argmax}_k P(y = k|x)$.

3.5 Training Procedure

BERT was fine-tuned using the HuggingFace Trainer API with: learning rate 2×10^{-5} , training batch size 32, evaluation batch size 64, 3 epochs, weight decay 0.01, and 500 warmup steps. These follow established defaults for BERT fine-tuning (Devlin et al., 2019). The model was optimised using cross-entropy loss:

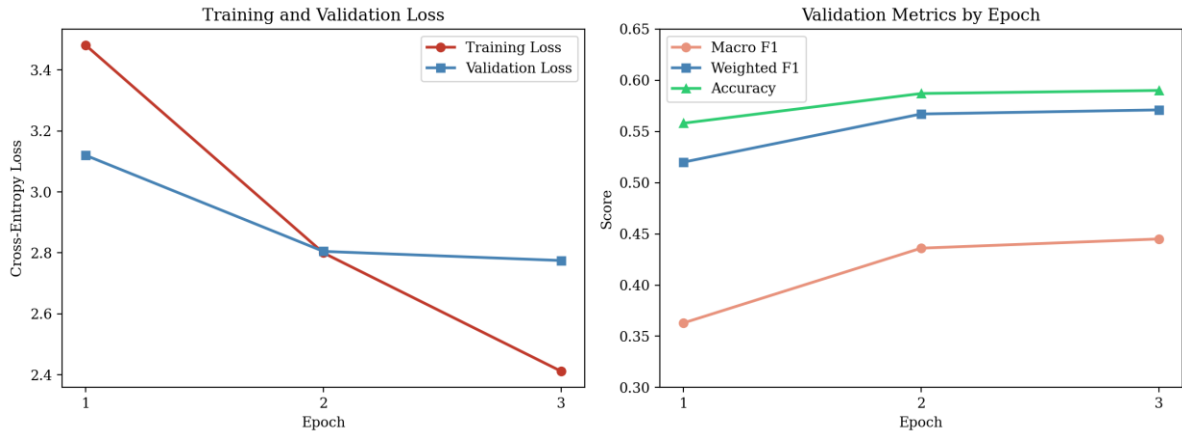
$$L = -\sum_k y_k * \log P(y = k | x),$$

where y_k is the ground-truth indicator for class k . Table 2 shows validation performance across the three training epochs. The largest improvement occurred between epoch 1 and epoch 2, where macro F1 increased from 0.363 to 0.436 (+7.3pp). By epoch 3, improvement was marginal (+0.9pp to 0.445), while validation loss decreased only slightly from 2.805 to 2.775. This pattern of diminishing returns, combined with the narrowing gap between training and validation loss, confirms three epochs as appropriate without overfitting risk.

Table 2: BERT validation performance across training epochs

Epoch	Train Loss	Val Loss	Accuracy	Macro F1	Weighted F1
1	3.481	3.120	0.558	0.363	0.520
2	2.800	2.805	0.587	0.436	0.567
3	2.412	2.775	0.590	0.445	0.571

Figure 2: Training/validation loss (left) and validation metrics (right) across epochs.



The model checkpoint with the highest weighted F1 on the validation set (epoch 3: 0.571) was selected as the final model. A fixed random seed of 42 was used throughout all experiments to ensure full reproducibility of results.

3.6 Evaluation Strategy

Stage 1 – In-Domain Evaluation.

Both models were evaluated on the GoEmotions held-out test set, comprising 5,427 examples that were not used during training or validation. Due to the significant class imbalance described in Section 3.2, weighted F1 was adopted as the primary evaluation metric. Weighted F1 computes the F1 score independently for each emotion class and then calculates a weighted average using class support as weights:

$$\text{Weighted } F_1 = \sum_k \frac{n_k}{N} * F_{1_k} ,$$

where n_k is support for class k and N is the total number of examples. Macro F1 (unweighted average across classes) and accuracy are also reported. Results were compared against the benchmark published by Demszky et al. (2020).

Stage 2 – Cross-Domain Evaluation and Manual Annotation.

Since the Twitter dataset lacks pre-existing emotion labels, a manual annotation process was required to create a gold-standard evaluation set. Both annotators independently labelled all 500 sampled tweets, each assigning a single dominant emotion from the 28-category GoEmotions taxonomy. To ensure consistency, annotators followed written guidelines specifying that: (1) only the dominant emotion should be selected; (2) neutral should be assigned only when genuinely no emotion was detected; and (3) where multiple emotions coexisted, annotators should select the one most likely to influence a customer service agent’s response priority. Inter-annotator agreement was measured using Cohen’s Kappa:

$$\kappa = \frac{p_o - p_e}{1 - p_e} ,$$

where p_o is the observed proportion of agreement and p_e is expected agreement by chance. We obtained $\kappa = 0.733$ with a raw agreement rate of 77.4%, falling within the substantial agreement range of 0.61 to 0.80 (Landis and Koch, 1977). Disagreements occurred most commonly on semantically adjacent pairs such as annoyance versus anger and disappointment versus disapproval, and were resolved through joint discussion to produce a consensus gold-standard label. The disagreement rate of 22.6% establishes a realistic performance ceiling: if trained annotators disagree on nearly a quarter of tweets, the model cannot be expected to achieve perfect accuracy.

Transfer reliability was assessed via pre-defined thresholds: strong ($\leq 10\%$ weighted F1 drop), moderate (10–20%), or poor ($> 20\%$). As a contingency, emotions were also grouped into broader classes (negative, positive, ambiguous) for re-evaluation.

3.7 Reproducibility

Consistent preprocessing, fixed random seeds, and publicly available datasets ensure reproducibility. All code is fully annotated and provided alongside this report.

4. Results

4.1 In-Domain Performance

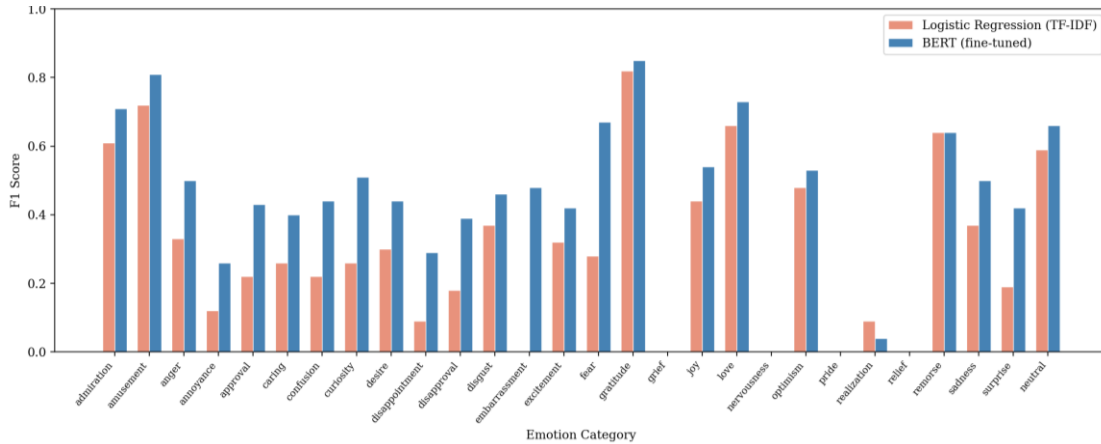
Table 3: In-domain performance on GoEmotions test set.

Model	Accuracy	Macro F1	Weighted F1
BERT-base (Demszky et al., 2020 benchmark)	—	0.461	—
Logistic Regression (TF-IDF)	0.503	0.305	0.446
BERT (fine-tuned, ours)	0.585	0.434	0.566

BERT outperforms the Logistic Regression baseline by 12.9 percentage points in macro F1, demonstrating the value of contextual representations for fine-grained emotion classification (Devlin et al., 2019; Harutyunyan and Kumar, 2026). Our BERT macro F1 (0.434) falls 2.7 points below the published benchmark (0.461), expected given that the benchmark used multi-label training whereas our pipeline applies single-label conversion.

Performance varied considerably across emotion classes (Figure 3). The model performed best on distinctive, well-represented emotions with clear lexical markers: gratitude (F1: 0.85), amusement (F1: 0.81), and love (F1: 0.73). Performance was near-zero on rare categories, nervousness (n=16), grief (n=6), relief (n=7), and pride (n=8), confirming that fine-grained classifiers are fundamentally constrained by class frequency and semantic overlap (Truong, 2024).

Figure 3: Per-emotion F1 scores for Logistic Regression and BERT on the GoEmotions test set.



4.2 Cross-Domain Transfer to Twitter

Table 4: Cross-domain transfer performance.

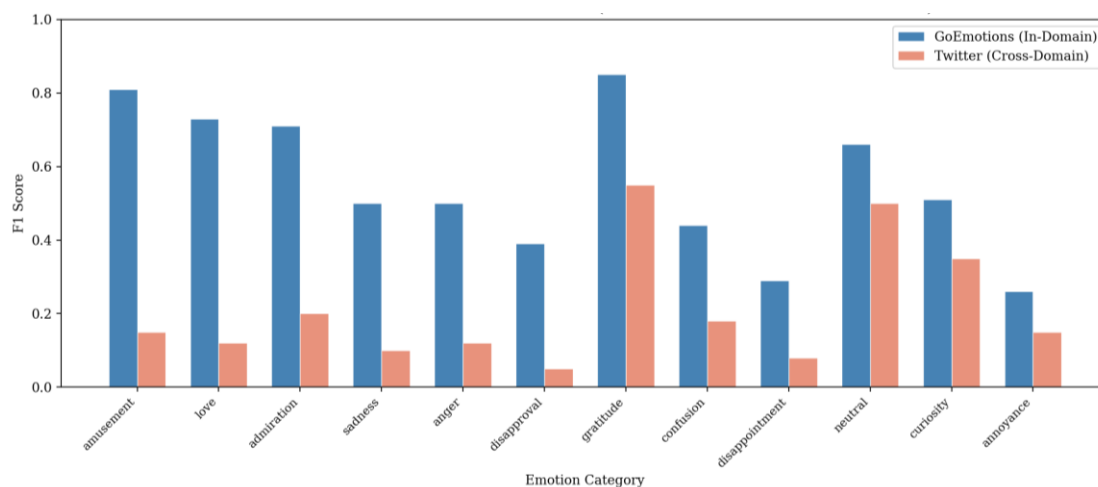
Model	Domain	Accuracy	Macro F1	Weighted F1	Transfer Gap
Logistic Regression	GoEmotions	0.503	0.305	0.446	—
Logistic Regression	Twitter	0.278	0.151	0.186	26.0pp
BERT	GoEmotions	0.585	0.434	0.566	—
BERT	Twitter	0.374	0.255	0.312	25.5pp

BERT achieved a weighted F1 of 0.312 on the annotated Twitter sample, a decline of 25.5 percentage points from its in-domain score. By our pre-defined thresholds, this constitutes poor transfer. The Logistic Regression baseline showed a near-identical gap of 26.0 percentage points. The similarity of these gaps is a critical finding: it indicates that the performance bottleneck is domain mismatch rather than model architecture, since both a context-free (TF-IDF) and a contextual (BERT) model degrade by approximately the same magnitude. These are point estimates on 500 tweets; bootstrap resampling could yield intervals of ± 3 –5pp, but the gap remains clearly within the poor transfer range.

4.3 Per-Emotion Transfer Analysis

To understand which emotions are most affected by domain shift, we compared BERT’s per-emotion F1 scores between GoEmotions and Twitter for the twelve emotions with sufficient support in the annotated sample (Figure 4). Transfer quality varies substantially across categories.

Figure 4: Per-emotion F1 comparison between GoEmotions (in-domain) and Twitter (cross-domain).



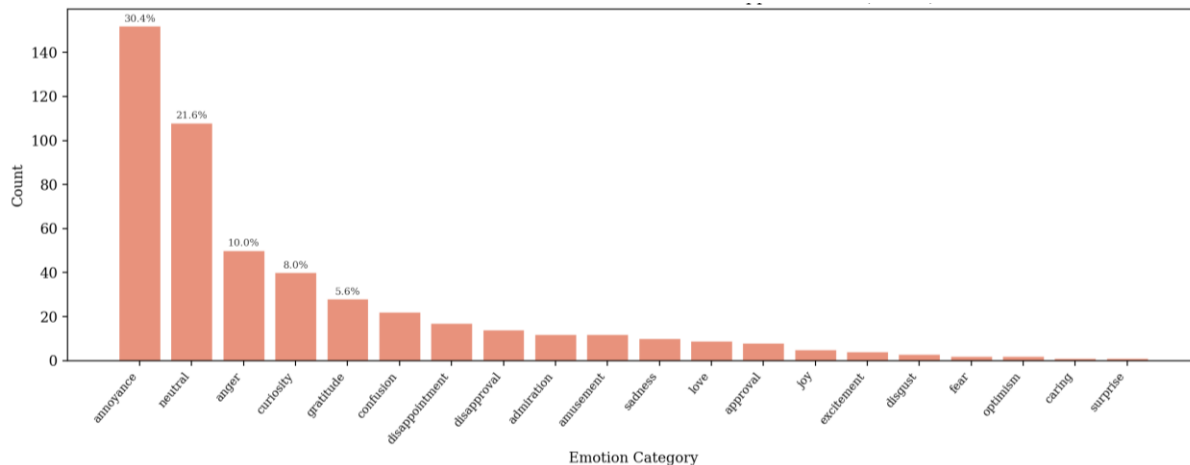
Gratitude showed the strongest transfer, declining only moderately from its in-domain F1 of 0.85. This is consistent with gratitude being expressed through distinctive, domain-invariant lexical markers (“thank you”, “thanks”, “appreciate”) that function similarly regardless of whether the context is Reddit or customer support. Neutral also transferred reasonably, likely because the model’s tendency to default to neutral under uncertainty coincidentally aligns with the high prevalence of neutral tweets in customer support.

By contrast, annoyance, the single most prevalent emotion in customer support at 30.4% of annotations, transferred very poorly. On Reddit, annoyance is expressed through explicit, direct language (“this is so annoying”), whereas in customer support it manifests as polite but firm complaints (“It’s going to cost me £42 to return this faulty item”). This mismatch is practically significant: the single most important emotion for customer service routing is the one the model handles worst. Disappointment and disapproval showed similarly large drops, confirming that negative emotions expressed through indirect, restrained language represent the primary failure mode in cross-domain transfer.

4.4 Emotion Distribution in Customer Support

Figure 5 shows the distribution of emotions in the 500 annotated customer support tweets. Annoyance (30.4%), neutral (21.6%), and anger (10.0%) collectively accounted for 62.0% of all human annotations. The dominance of annoyance over anger is notable: customer support frustration is more commonly expressed as low-intensity irritation than overt hostility, a distinction that has implications for how customer service teams should prioritise responses. The most common positive emotions were gratitude (5.6%) and admiration (2.4%), typically occurring in tweets acknowledging a resolution or expressing appreciation for help received.

Figure 5: Emotion distribution in annotated customer support tweets ($n=500$).



When BERT’s predictions on the full 10,000-tweet sample are examined, neutral dominates at 51.8%. However, given the model’s 62.6% error rate on annotated data, this predicted distribution is approximate and likely reflects the model defaulting under domain uncertainty.

4.5 Grouped Emotion Analysis

Table 5: Fine-grained vs. grouped classification on Twitter.

Scheme	Weighted F1 (Twitter)	Improvement
Fine-grained (28 classes)	0.312	—
Grouped (negative, positive, ambiguous)	0.485	+17.3pp

Collapsing 28 emotions into three broader groups improved weighted F1 by 17.3 percentage points. Figure 6 shows the grouped confusion matrix, which reveals an important asymmetry in the model’s behaviour. The model achieves high precision for negative emotions (0.90) but very low recall (0.24): when it predicts a tweet as negative, it is almost always correct, but it misses the large majority of truly negative tweets, predominantly misclassifying them as ambiguous. This pattern has direct practical implications for deployment, discussed further in Sections 5.4 and 7.

Figure 6: Grouped emotion confusion matrix. Most negative tweets are misclassified as ambiguous.

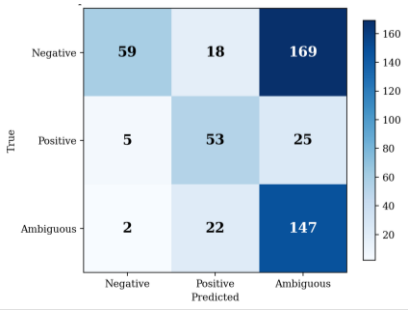
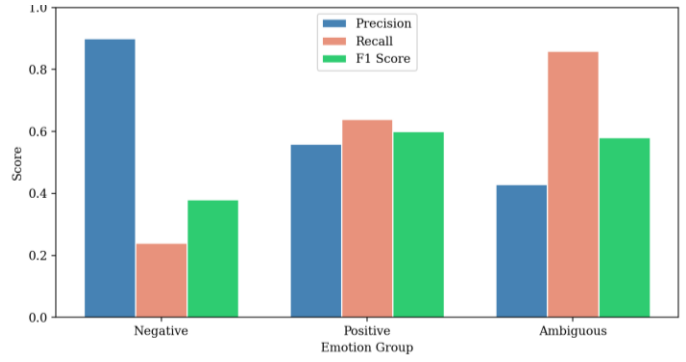


Figure 7: Precision, recall, and F1 for grouped emotion classification (BERT on Twitter).

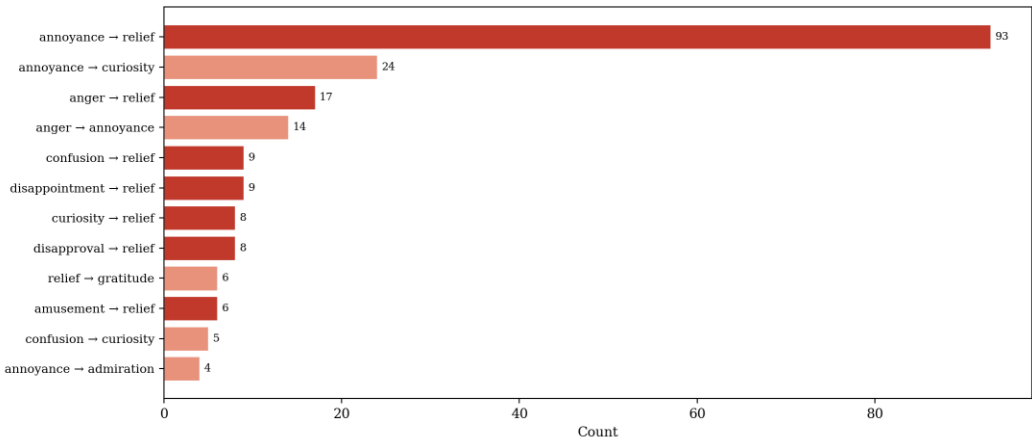


5. Error Analysis and Discussion

5.1 Systematic Misclassification Patterns

The model misclassified 313 of 500 annotated tweets (62.6%), substantially higher than the in-domain error rate of 41.5% on GoEmotions. Errors were not randomly distributed but clustered around specific emotion pairs (Figure 8). The most striking pattern was the model’s tendency to predict relief for tweets annotated as annoyance, accounting for 93 cases alone. Anger was also frequently predicted as relief (17 cases) or annoyance (14 cases), while confusion and disappointment were each mislabelled as relief multiple times. Beyond this anomaly, semantically adjacent pairs such as anger/annoyance and approval/gratitude were frequently confused, reflecting the inherent overlap in the GoEmotions taxonomy.

Figure 8: Top misclassified emotion pairs on Twitter. Red bars highlight relief-related errors.



5.2 Qualitative Error Analysis

Table 6 presents representative misclassified tweets illustrating the three primary failure modes. These examples demonstrate that errors follow systematic patterns tied to differences between Reddit and customer support language, rather than occurring randomly.

Table 6: Representative misclassified tweets with identified failure modes.

Tweet (abbreviated)	True	Predicted	Failure Mode
Thanks, but it’s been nearly 48 hours and it still says membership will be active soon	annoyance	gratitude	Polite framing
It’s going to cost me £42 to return this faulty item. Are you going to cover this?	annoyance	curiosity	Indirect emotion
Now I can’t download any apps, free or otherwise.	annoyance	neutral	Event description
When I pay £417 for my season ticket, I expect you to get me to London and back.	anger	neutral	Indirect anger
I’m switching to [competitor]	annoyance	neutral	Implied frustration
Ok.. I wish [brand] fix this darn phone.	annoyance	desire	Lexical mismatch
[Brand] is useless. They can’t help me on the phone. #stupid	anger	annoyance	Semantic overlap

The polite framing failure is particularly consequential for customer support applications. Customers frequently combine polite language with complaint content (“thanks, but...”), a pragmatic convention largely absent from Reddit where frustration is expressed more directly. The model, having learned that “thanks” reliably correlates with gratitude on Reddit, applies this association even when surrounding context clearly signals annoyance. Similarly, tweets expressing emotion through event description alone (“I can’t download any apps”) lack explicit affective vocabulary, causing the model to default to neutral. These patterns suggest the model learned surface-level lexical associations rather than deeper pragmatic understanding of emotional communication in different contexts.

5.3 The Relief Anomaly and Model Calibration

Relief is among the rarest GoEmotions classes ($n=96$ training examples, $F1: 0.00$ in-domain), yet it dominates cross-domain errors with 93 cases of annoyance predicted as relief alone. The model’s average prediction confidence on Twitter was 0.654, lower than in-domain, indicating less peaked softmax distributions on out-of-distribution text. Under this uncertainty, the classification head over-activates a poorly learned class, producing systematically implausible labels that would be directionally misleading in production, potentially routing frustrated customers away from escalation.

5.4 Discussion

The central finding of this study is that fine-grained emotion detection transfers poorly from general social media to customer support. BERT improves substantially over Logistic Regression in-domain, yet both models show near-identical transfer gaps of approximately 25–26 percentage points in weighted F1. This convergence indicates that the bottleneck is the domain itself, customer support language is shorter, more transactional, more polite, and more indirect than

Reddit, rather than model architecture. A more powerful model cannot compensate for training data that does not represent the target distribution (Ruder, 2019).

The per-emotion analysis in Section 4.3 reveals that this domain gap is not uniform across categories. Emotions with domain-invariant lexical markers, such as gratitude and neutral, transfer reasonably well, while emotions whose expression depends on pragmatic context, such as annoyance and disappointment, degrade severely. This finding has methodological implications: domain adaptation efforts should target negative emotions expressed through indirect language specifically, rather than attempting uniform improvement across all 28 categories.

The grouped evaluation demonstrates that broad emotional valence transfers more reliably, with weighted F1 rising from 0.312 to 0.485 when labels are collapsed into three groups. However, the grouped confusion matrix (Figure 6) reveals an important practical asymmetry: the model achieves 0.90 precision but only 0.24 recall for negative emotions. In a production routing system, the model could reliably confirm negative emotion when it detects it, but would miss the majority of frustrated customers, requiring human oversight to supplement automated triage.

We considered whether methodological artefacts, single-label conversion or label ordering bias could partly explain the gap. However, these equally affect in-domain performance, where BERT’s 12.9pp improvement over Logistic Regression confirms meaningful learning. The transfer gap is therefore best attributed to genuine domain shift.

Future work should explore domain-adaptive pre-training on unlabelled customer support text, which Gururangan et al. (2020) demonstrated can substantially improve performance in specialised domains and which our findings suggest would be more effective than simply scaling to a larger model. Multi-label classification, leveraging conversational context across message threads, and psycholinguistic feature augmentation (Zanwar et al., 2022) represent additional promising directions for closing the transfer gap.

6. Limitations

Several limitations should be considered when interpreting our results. First, the single-label conversion discards co-occurring emotions, and the label ordering in GoEmotions reflects annotator sequence rather than emotion dominance. This may bias the training signal, though its impact is mitigated by the fact that 86% of GoEmotions examples carry only a single label.

Second, our evaluation set of 500 annotated tweets provides limited statistical power for per-emotion analysis on rare categories. Emotions such as fear ($n=2$), caring ($n=1$), and surprise ($n=1$) have too few examples for reliable individual conclusions, meaning our per-emotion transfer analysis is necessarily limited to the twelve most frequent categories.

Third, our data sources carry platform-specific biases. Twitter’s 280-character limit constrains emotional expression differently from other support channels such as email, live chat, or phone transcripts. Results may not generalise to these longer-form contexts where customers have more space to articulate emotions. Additionally, the Twitter dataset dates from 2017; customer support language and platform conventions may have evolved, and this temporal gap should be considered when applying our findings.

Fourth, we identified domain mismatch as the primary bottleneck but did not attempt domain-adaptive pre-training or few-shot fine-tuning, which could demonstrate whether modest in-domain data can close the gap. This remains the most important direction for future work.

7. Conclusion and Recommendations

This paper asked whether an NLP model trained on social media emotions can reliably identify fine-grained emotions in customer support messages. The answer is: only partially. Fine-tuned BERT performed well in-domain (*weighted F1* = 0.566) but dropped sharply on Twitter customer support data (*weighted F1* = 0.312). The transfer gap, together with the qualitative error patterns identified in Section 5, demonstrates that a model trained on Reddit cannot be assumed to generalise to professional support interactions without domain-specific adaptation.

Three recommendations follow from these findings:

First, organisations should avoid deploying fine-grained emotion labels for automated decision-making in customer support until domain-specific adaptation has been undertaken. The relief anomaly documented in Section 5.3 illustrates how uncalibrated predictions can produce systematically implausible outputs that would actively undermine customer experience rather than improve it.

Second, grouped emotion labels (negative, positive, ambiguous) provide a realistic basis for immediate deployment. Our analysis showed 0.90 precision for negative emotions, when the model flags frustration, it is almost always correct. A practical architecture would use this as a first-pass filter, routing high-confidence negative messages to experienced agents while remaining messages follow standard queuing. Since annoyance accounts for 30.4% of customer emotions, even partial detection could meaningfully reduce resolution times.

Third, the most effective next step is collecting a modest in-domain labelled dataset. Our annotation process demonstrated that labelling 500 tweets with substantial inter-annotator agreement ($\kappa = 0.733$) is practically achievable with two annotators following clear guidelines. Since domain mismatch rather than model size is the core constraint, even a small in-domain training corpus could substantially improve performance through domain-adaptive fine-tuning, without requiring a larger or more expensive model.

For customer support teams, human emotional intelligence remains essential in service interactions. Agents should be trained to recognise the subtle, indirect emotional cues, polite frustration, implied disappointment, restrained anger that current models consistently miss. Our analysis of the emotion distribution suggests that de-escalation training should be prioritised, given that annoyance and anger together account for over 40% of customer emotions. The most effective customer support strategy combines automated coarse-grained routing with human fine-grained emotional response.

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Declaration of AI Usage

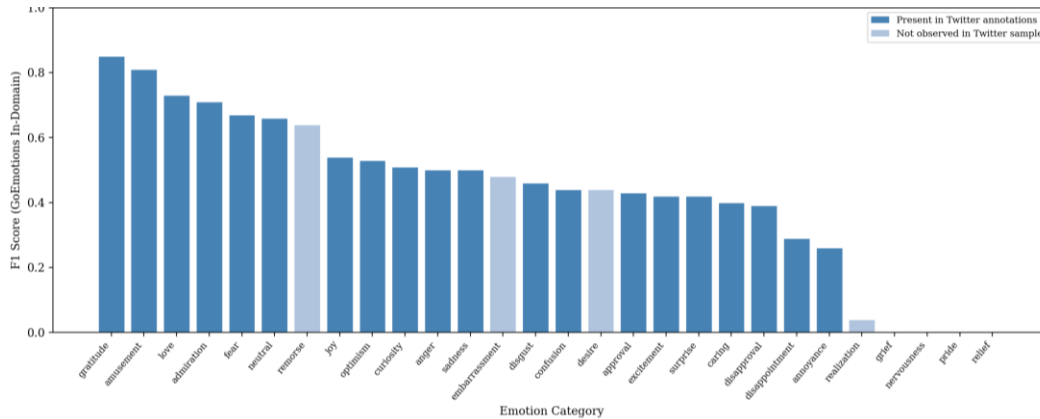
In accordance with UCL’s policy on the assistive use of artificial intelligence tools, we declare the following uses of generative AI during this project: Claude (Anthropic) was used in an assistive capacity for debugging code, formatting the report document, ChatGPT (OpenAI) was used for proofreading and grammar checking of draft text. All substantive intellectual contributions like research design, data analysis, interpretation of results, and written argumentation are entirely our own. The annotation of 500 tweets was performed manually by two team members without AI assistance. All AI-generated suggestions were critically reviewed and modified where necessary.

Appendix

All code, data, and annotation files are provided in the accompanying Jupyter Notebook (GP_team 1.ipynb).

Appendix A: BERT In-Domain F1 by Emotion

Figure A1: BERT in-domain F1 by emotion. Darker bars = emotions observed in Twitter annotations.



Appendix B: Annotation Guidelines

Both annotators independently labelled all 500 sampled tweets. Guidelines: (1) read each tweet and assign one dominant emotion from the 28 GoEmotions categories; (2) if multiple emotions were present, choose the most prominent; (3) label as neutral only if genuinely no emotion was detected; (4) where multiple emotions coexist, select the one most likely to influence a customer service agent's response. Disagreements were resolved through joint discussion.

Appendix C: Member Contributions

Member	Contribution
Leya Sherman	BERT fine-tuning, manual annotation (250 tweets), methodology writing
Marcus Hudson	Related works, error analysis writing
Majid Ahmadi Moghaddam	Exploratory data analysis, results visualisations
Pablo Williams	Data preprocessing pipeline, grouped emotion evaluation
Salman Khalid	TF-IDF baseline, manual annotation (250 tweets), cross-domain evaluation, results writing